



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic decisions.

Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Values

Review Rating only

Demographics

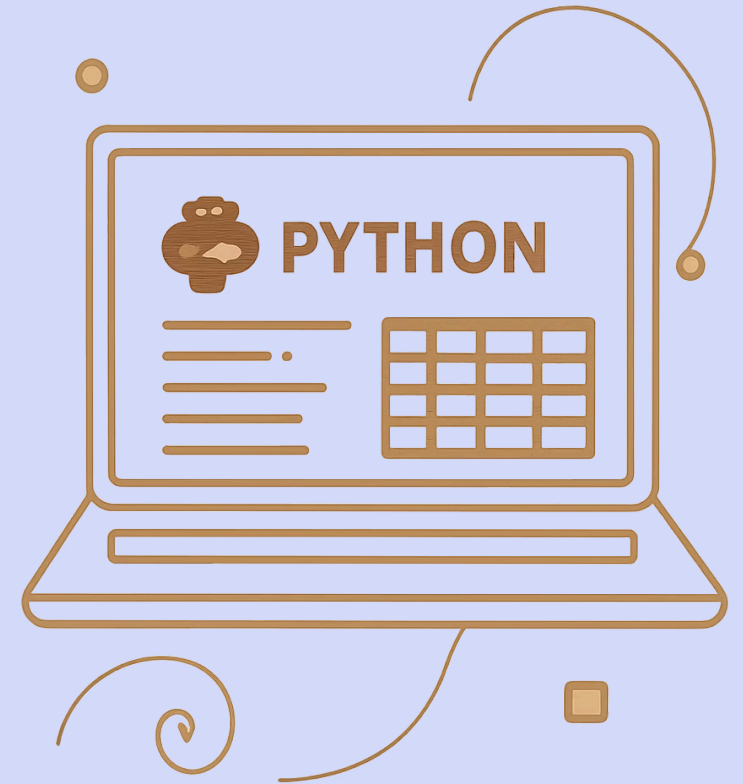
Age, Gender, Location, Subscription Status

Purchase & Behavior

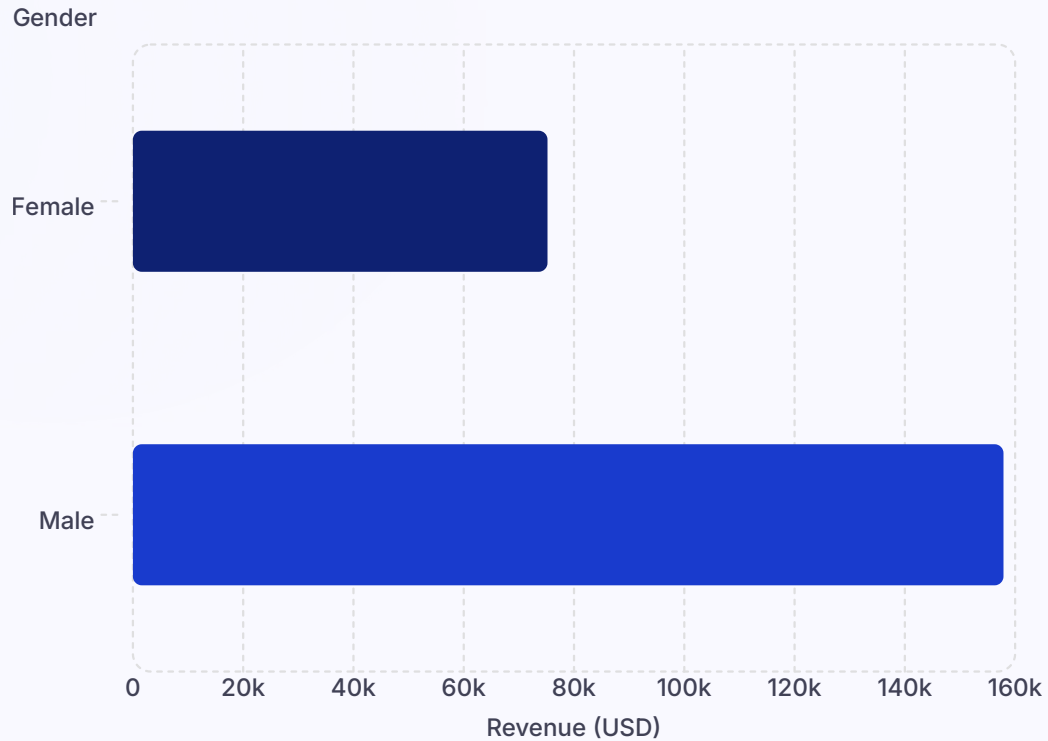
Category, Amount, Discount, Shipping, Review Rating

Data Preparation in Python

- **Clean & Standardize**
Imputed missing Review Ratings by category median; renamed columns to snake_case
- **Feature Engineering**
Created **age_group** bins and **purchase_frequency_days**
- **Load to Database**
Cleaned DataFrame loaded into PostgreSQL for SQL analysis



Revenue by Gender

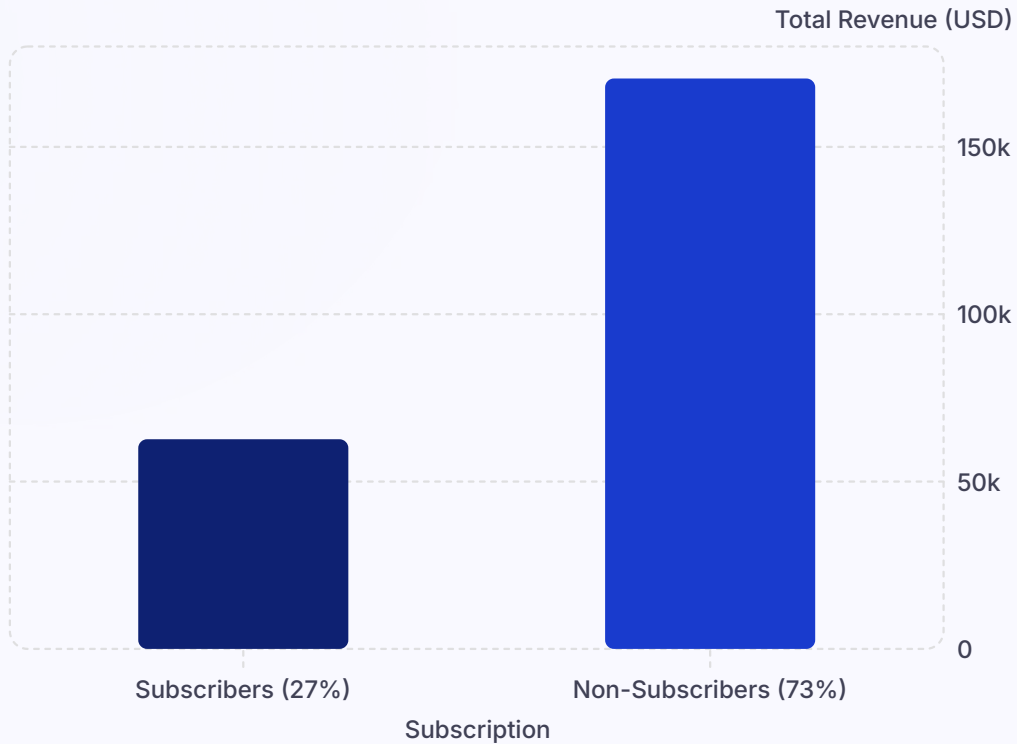


Key Insight

Male customers generated **2.1× more revenue** than female customers across all categories.

Male: \$157,890 · Female: \$75,191

Subscribers vs. Non-Subscribers



Spending Patterns

Average spend is nearly identical: **\$59.49** (subscribers) vs **\$59.87** (non-subscribers).

Non-subscribers drive **73%** of total revenue.

Products & Discounts

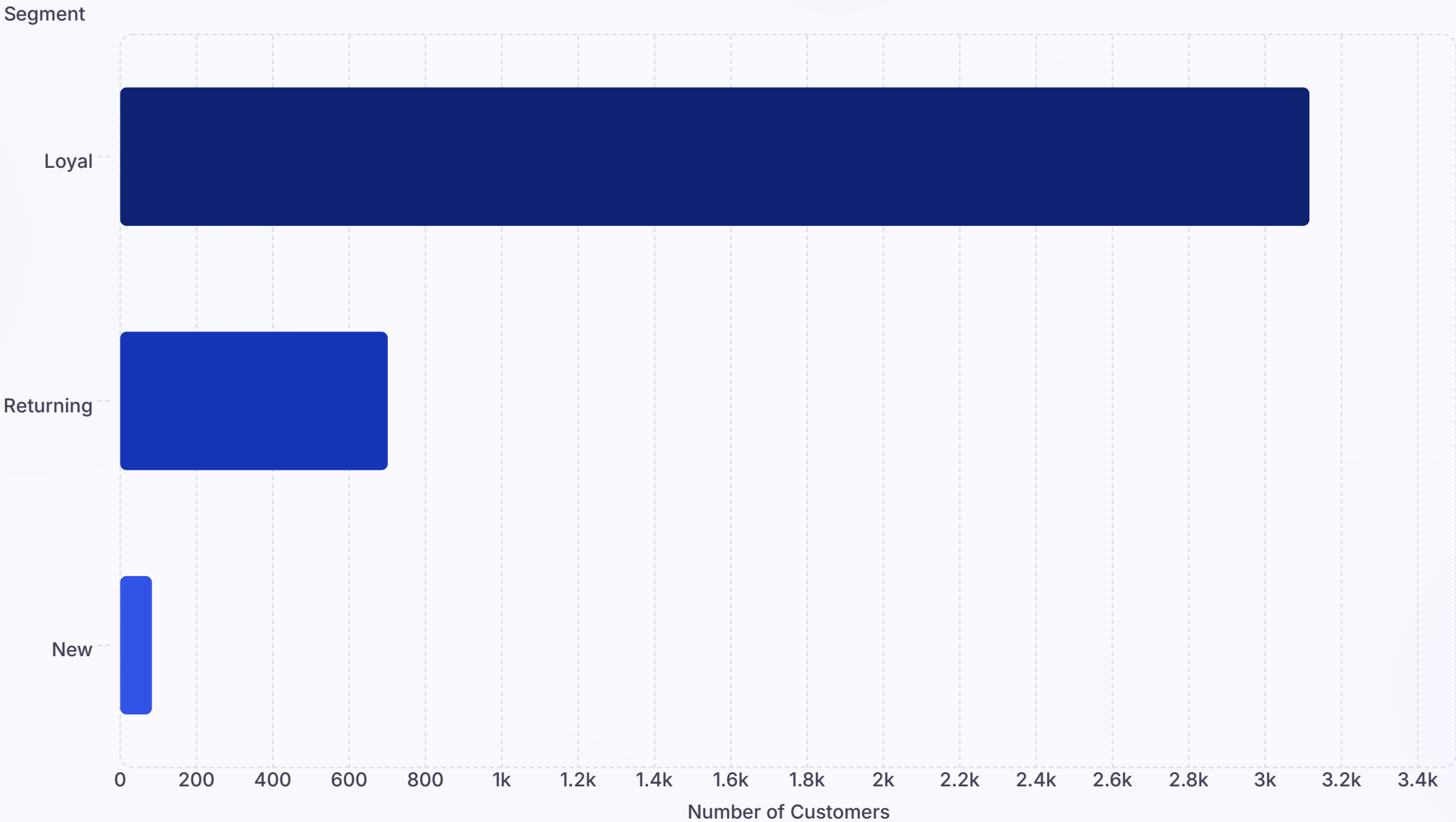
Top 5 Products by Rating

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

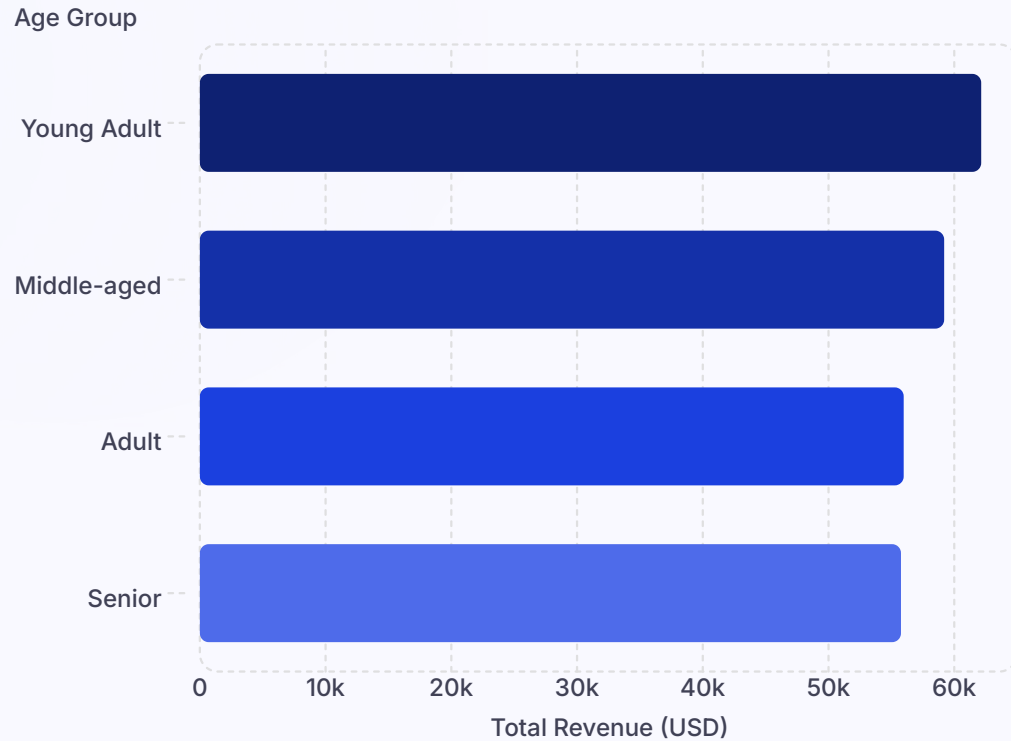
Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation



80% of customers are classified as Loyal (based on purchase history). Only 83 are New customers – a growth opportunity.

Revenue by Age Group



Key Takeaway

Young Adults lead in revenue contribution. Revenue is relatively evenly distributed across all age groups.

Power BI Dashboard



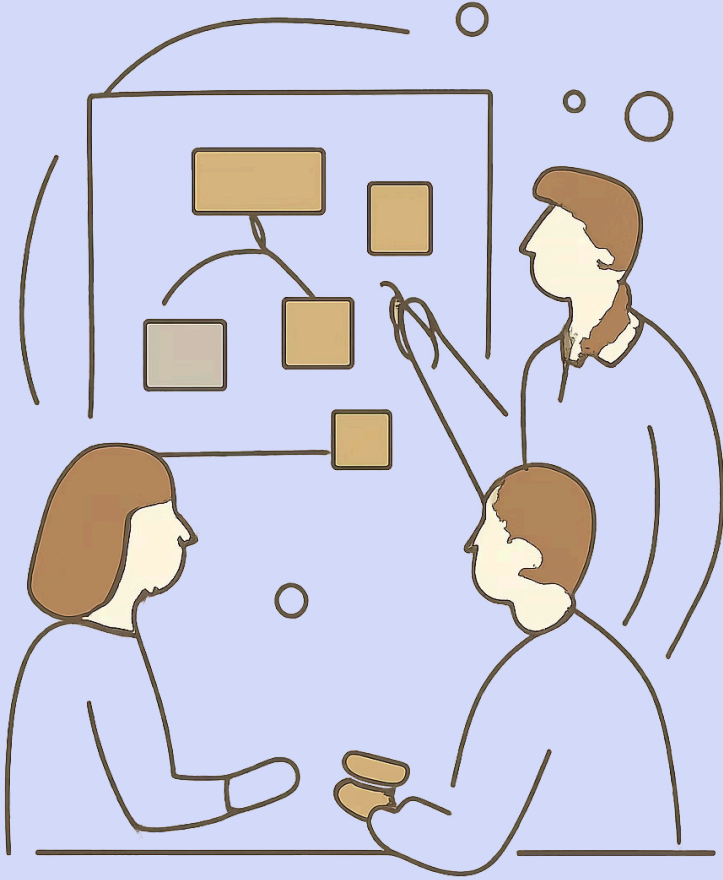
Interactive Insights

Visual breakdowns by Category, Age Group, Subscription Status, and Shipping Type.

3.9K
Customers

\$59.76
Avg Purchase

3.75
Avg Rating



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow the Loyal segment



Review Discounts

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users